Eduvos Assessment Coversheet

Academic Details

Campus:	**EDUVOS ONLINE**
Faculty:	**INFORMATION TECHNOLOGY**
Module Code:	**ITBDA4-14**
Lecturer's Name:	**MR LEKULA.G**
Group (if applicable)	

Student Details (Individual, if applicable)

Student Full Name: **DANIELLA MARGUAX MYBURGH**

Student Number: **EDUV1598917**

Academic Integrity

AI Disclosure Completed [X] **Yes** [ ] **No**

Declaration:
I declare that this assessment is my own original work except for source material explicitly acknowledged. I confirm that this work has not previously been submitted for assessments in any course. I acknowledge the institutional AI policy and confirm that AI tools have only been used as stipulated and documented in the AI Disclosure Appendix.

Student Signature 

Date **2026/11/02**

Lecturer Use Only

Lecturer Comments:

Marks Awarded (%):

Lecturer Signature:

Date:

Table of Contents

EDUVOS ASSESSMENT COVERSHEET 1

ASSESSMENT 4

QUESTION 1: 4
QUESTION 2: 6

QUESTION 3: 7

QUESTION 4: 8

WORKS CITED 9

AI DISCLOSURE APPENDIX 11

DECLARATION: 11

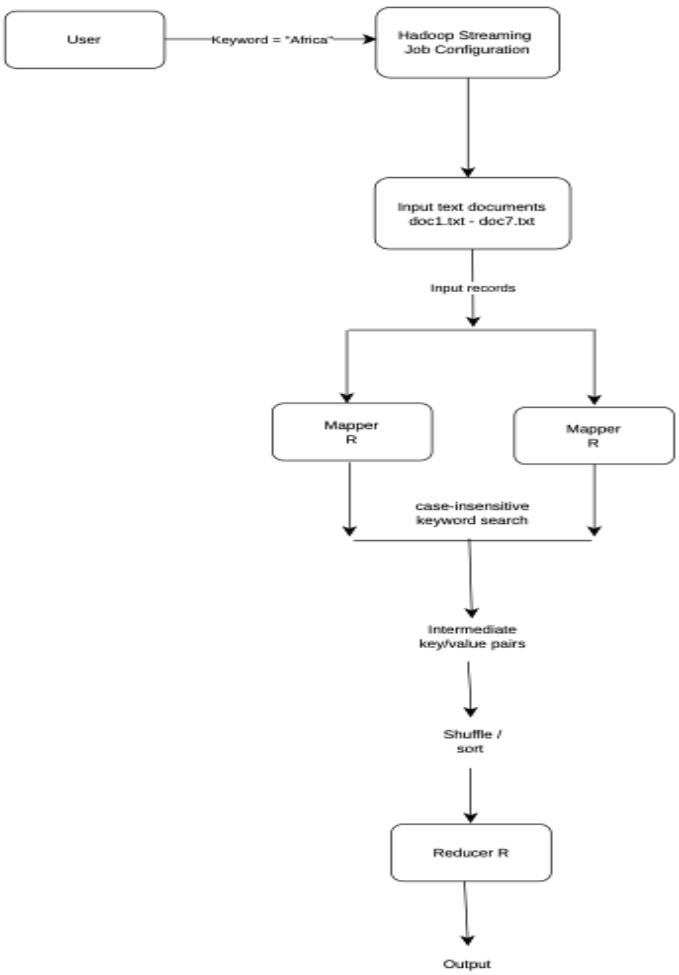
Assessment

Question 1:

1.1)

MapReduce processes large datasets by dividing computation into two principal phases: Map and Reduce. The map function processes input records and generates intermediate key-value pairs, while the reduce function receives values grouped by their intermediate key and produces the final output. (Taylor, 2010)

For this task, the input consists of the supplied news documents. The user supplies a keyword dynamically, such as "Africa". Each Mapper searches its assigned document content case-insensitively and records the document name, row and column of every exact match. (Taylor, 2010)



https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=Question_1_ITBDA4-14_EDUV1598917.drawio&dark=auto#R%3Cmxfile%3E%3Cdiagram%20name%3D%22Page-

1%22%20id%3D%22r3ePUSQmL3wtRzq_4cr6%22%3E7Vrfl6I2FP5rfJw9QPjl4

The process is the documents are supplied to Hadoop as input. The keyword is entered dynamically at runtime. The R Mapper reads each record and searches for the keyword without regard to case.Every match produces an intermediate key-value pair where the key is the filename and the value is the row and column location. Hadoop performs shuffle and sort, grouping Mapper output by filename. The R Reducer writes the grouped matches into the final result. This use of MapReduce is appropriate because the search can be divided into independent pieces and processed in parallel. Dean and Ghemawat note that MapReduce programs can automatically be parallelised across multiple machines while the framework handles partitioning, scheduling and communication. (Taylor, 2010)

1.2) (A map-reduce method to iterate blocks of file-array data with...) (*Map reduce — internals. 1) introduction | by Mandeep Singh | Distributed-Systems | Medium*)

1.3) (*Understanding reducers from scratch | by Arseniy Tomkevich | Medium*) (Mild, *Reduce() – apply's lesser known brother: R-bloggers* 2017)

1.4)

Question 2:

2.1) (Nugent, Debar - details of the denoising algorithm) (GeeksforGeeks, Data cleaning in R 2026)

2.2) (Yan, Notes for "text mining with R: A tidy approach" 2020) (*Beginner's guide to LDA topic modelling with R | by Farren Tang | TDS Archive | Medium*)

2.3) (Beginner's guide to LDA topic modelling with R | by Farren Tang | TDS Archive | Medium) (GeeksforGeeks, Data visualization in R 2026)

Question 3:

3.1) (GeeksforGeeks, Web scraping using R language 2025) (*Web scraping in R*)

3.2) (Contributed by Roshini Sudhaharan, Text pre-processing in R) (*Data cleaning in R made simple | towards data science*)

3.3) (A comprehensive guide to data visualisation in R for beginners | towards data science) (Mastering data visualization in R: A comprehensive guide to creating stunning visuals and communicating data insights | by NILIMESH Halder, Phd | medium)

Question 4:

4.1) (Hadley Wickham, *24 web scraping*)

4.2) (Blog, *Best practices for data cleaning and preprocessing: R-bloggers* 2023)

4.3) (GeeksforGeeks, *PostgreSQL - import CSV file into table* 2025)

4.4) (*How can I make my github repository open source and accessible to the public? · community · discussion #170965*)

Works Cited

1. Taylor, R.C. (2010) *An overview of the Hadoop/mapreduce/hbase framework and its current applications in bioinformatics - BMC bioinformatics*, *SpringerLink*. Available at: https://link.springer.com/article/10.1186/1471-2105-11-s12-s1 (Accessed: 02 November 2026).
2. *A map-reduce method to iterate blocks of file-array data with...* (no date) *R*. Available at: https://search.r-project.org/CRAN/refmans/filearray/html/mapreduce.html (Accessed: 02 November 2026).
3. (No date) *Map reduce — internals. 1) introduction | by Mandeep Singh | Distributed-Systems | Medium*. Available at: https://medium.com/mitdistributedsys/map-reduce-internals-98e4eec30a3c (Accessed: 02 November 2026).
4. (No date a) *Understanding reducers from scratch | by Arseniy Tomkevich | Medium*. Available at: https://medium.com/@jsmuster/understanding-reducers-d0f934aceccd (Accessed: 02 November 2026).
5. Mild (2017) *Reduce() – apply's lesser known brother: R-bloggers*, *R*. Available at: https://www.r-bloggers.com/2017/07/r-reduce-applys-lesser-known-brother/ (Accessed: 02 November 2026).
6. Nugent, C.M. (no date) *Debar - details of the denoising algorithm*. Available at: https://cran.r-project.org/web/packages/debar/vignettes/debar-algorithm-details.html (Accessed: 02 November 2026).
7. GeeksforGeeks (2026) *Data cleaning in R*, *GeeksforGeeks*. Available at: https://www.geeksforgeeks.org/r-language/data-cleaning-in-r/ (Accessed: 02 November 2026).
8. Yan, Q. (2020) *Notes for 'text mining with R: A tidy approach'*, *6.1 Latent Dirichlet Allocation*. Available at: https://bookdown.org/Maxine/tidy-text-mining/latent-dirichlet-allocation.html (Accessed: 02 November 2026).
9. (No date a) *Beginner's guide to LDA topic modelling with R | by Farren Tang | TDS Archive | Medium*. Available at: https://medium.com/data-science/beginners-guide-to-lda-topic-modelling-with-r-e57a5a8e7a25 (Accessed: 02 November 2026).
10. GeeksforGeeks (2026b) *Data visualization in R*, *GeeksforGeeks*. Available at: https://www.geeksforgeeks.org/r-language/data-visualization-in-r/ (Accessed: 02 November 2026).
11. GeeksforGeeks (2025) *Web scraping using R language*, *GeeksforGeeks*. Available at: https://www.geeksforgeeks.org/r-language/web-scraping-using-r-language/ (Accessed: 02 November 2026).
12. *Web scraping in R* (no date) *Stats and R*. Available at: https://statsandr.com/blog/web-scraping-in-r/ (Accessed: 02 November 2026).
13. Contributed by Roshini Sudhaharan (no date) *Text pre-processing in R*, *Tilburg Science Hub*. Available at: https://www.tilburgsciencehub.com/topics/manage-manipulate/manipulate-clean/textual/text-preprocessing/ (Accessed: 02 November 2026).
14. (No date) *Data cleaning in R made simple | towards data science*. Available at: https://towardsdatascience.com/data-cleaning-in-r-made-simple-1b77303b0b17/ (Accessed: 02

November 2026).

15. (No date a) *A comprehensive guide to data visualisation in R for beginners | towards data science*. Available at: https://towardsdatascience.com/a-guide-to-data-visualisation-in-r-for-beginners-ef6d41a34174/ (Accessed: 02 November 2026).
16. (No date a) *Mastering data visualization in R: A comprehensive guide to creating stunning visuals and communicating data insights | by NILIMESH Halder, Phd | medium*. Available at: https://medium.com/@HalderNilimesh/mastering-data-visualization-in-r-a-comprehensive-guide-to-creating-stunning-visuals-and-f733564a8a41 (Accessed: 02 November 2026).
17. Hadley Wickham, M.Ç.-R. (no date) *24 web scraping*, *24 Web scraping – R for Data Science (2e)*. Available at: https://r4ds.hadley.nz/webscraping.html (Accessed: 04 November 2026).
18. Blog, T.J.R. (2023) *Best practices for data cleaning and preprocessing: R-bloggers*, *R*. Available at: https://www.r-bloggers.com/2023/08/best-practices-for-data-cleaning-and-preprocessing/ (Accessed: 04 November 2026).
19. GeeksforGeeks (2025) *PostgreSQL - import CSV file into table*, *GeeksforGeeks*. Available at: https://www.geeksforgeeks.org/postgresql/postgresql-import-csv-file-into-table/ (Accessed: 04 November 2026).
20. *How can I make my github repository open source and accessible to the public? · community · discussion #170965* (no date) *GitHub*. Available at: https://github.com/orgs/community/discussions/170965 (Accessed: 04 November 2026).
21. Blei, D.M., Ng, A.Y. and Jordan, M.I. (2003) 'Latent Dirichlet Allocation', Journal of Machine Learning Research, 3, pp. 993–1022.
22. EMC Education Services (2015) Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data. Indianapolis: John Wiley & Sons.

AI Disclosure Appendix

AI tool used	Purpose / how it was used	How I verified or adapted the output
e.g. ChatGPT 4o	e.g. Brainstormed initial argument outline	e.g. Cross-checked with three peer-reviewed sources; fully rewritten in own words
e.g. Claude Opus 4.7	e.g. Brainstormed initial argument outline	e.g. Cross-checked with three peer-reviewed sources; fully rewritten in own words
e.g. Grok 4	e.g. Brainstormed initial argument outline	e.g. Cross-checked with three peer-reviewed sources; fully rewritten in own words

Declaration:

I/we declare that this assessment is my/our own original work, except for source material explicitly acknowledged. I/we confirm that any use of Artificial Intelligence (AI) tools has been applied only as permitted by the assessment instructions and has been fully disclosed where required.

I/we understand that I/we remain responsible for the accuracy, originality, and academic quality of this submission. I/we am/are aware of the institution's policies on academic integrity and the requirements relating to the responsible use of AI in assessments.

I/we acknowledge that failure to comply with these requirements, including undeclared or inappropriate AI use, may result in academic penalties or disciplinary procedures in accordance with institutional policy.

Student Signature(s)	Date